Sai Ganesh Takkellapati

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EDUCATION

University of New Haven, Connecticut **New Haven, CT**
Master of Science, Computer Science 2023 – Expected Dec 2024
Coursework: *Database systems, Artificial intelligence, Windows Network Administration, IOT, Computer security , Enterprise Network Design Web Database Development, Data Mining, Java Programmimg, Script Programming/Python .*

Sathyabama Institute of Science and Technology **Chennai, India**
Bachelor of Technology, Computer Science and Engineering July 2019 – May 2023
Coursework: *Foundation of Data Analytics, Data Visualization, Natural Language Processing, Machine Learning, statistical data analysis*

TECHNICAL SKILLS

Programming: SQL, Python, Java, JavaScript
Data Analysis: Power BI, Databricks, PySpark, R
Databases & Cloud: Azure, AWS, MySQL, MongoDB, PostgreSQL
Tools & Frameworks: Git, Figma, Docker
Web Technologies: HTML5, CSS, Bootstrap, FastAPI, Django, Flask

WORK EXPERIENCE

University of New Haven ,CT
Student Researcher [Python, PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas] Jan 2024 - Present

- Research on improving energy efficiency in large language models, like GPT-2, using low-rank factorization, distillation, quantization methods was carried out.
- Research was done on how to use distillation, quantization, and low-rank factorization to improve the energy efficiency of big language models such as GPT-2.
- Distillation, quantization, and low-rank factorization have all been studied as ways to increase the energy efficiency of large language models like GPT-2.
- In order to optimize machine learning models, I led a research program that entailed working with peers and faculty. This involved effective project management and cross-disciplinary communication.

HCL Technologies LTD
Web Development Intern [HTML, CSS, JAVASCRIPT]
Sathyabama University, Chennai, INDIA Jan 2023–May 2023

- Developed and maintained responsive websites and web applications.Worked on front-end development, ensuring cross-browser compatibility
- Led initiatives to promote HCL Technologies on campus, driving student engagement through seminars and workshops on cutting-edge advancements.
- Integrated APIs and worked with databases to fetch and display data dynamically.

ACADEMIC PROJECTS

Social Distancing & Face Mask Detection System with Real-Time Alert Integration [Python| CNN| YOLO V3 | Open CV | Machine Learning |Tensor Flow | Keras | Mobile Net]
Recognized as an Innovative Idea in Final Year Undergraduate Project

- Engineered a sophisticated deep learning solution for real-time detection of face mask usage and monitoring of social distancing in public spaces.
- Leveraged advanced machine learning techniques, particularly Convolutional Neural Networks (CNN) using TensorFlow and Keras, to accurately classify individuals as masked or unmasked.
- Integrated the system with existing surveillance infrastructure, employing OpenCV for efficient image and video processing to support large-scale, real-time monitoring.
- Implemented an automated alert mechanism that triggers alarms upon detection of non-compliance, enhancing public safety and supporting COVID-19 mitigation efforts.
- Deployed machine learning models using AWS SageMaker, ensuring scalability and efficient cloud integration for real-time inference

Online Quiz Application [Angular JS | HTML | CSS |Java script]

- Designed and developed a fully functional quiz application using HTML, CSS, and JavaScript.
- Implemented features such as multiple-choice questions, score calculation, and dynamic content updates.
- Created an intuitive user interface to enhance user engagement.Improved code efficiency by implementing clean coding practices and optimizing JavaScript functions.

Geocoding and Mapping with Python Using Data Sceince [| Python | Google Map API | Pandas]

- Processed and cleaned the dataset of universities for geospatial mapping.
- Implemented geocoding techniques to convert university addresses into geographical coordinates (latitude and longitude).
- Visualized the university locations on an interactive map using Python libraries like Folium.
- Enhanced user experience by making the map interactive and easy to navigate.

CERTIFICATIONS & PUBLICATIONS

- **Sai Ganesh,** Rizwan, Velvizhi. “Face Mask Detection with Alarm using Machine Learning.” **IRMJETS** https://www.irjmets.com/uploadedfiles/paper//issue_3_march_2023/34706/final/fin_irjmets1679571000.pdf
- Python Using Data Science (IRMS)
- Web development Intern Certificate (HCL Technologies)